EGR 105 Lab 5: Cantilever Beam Report

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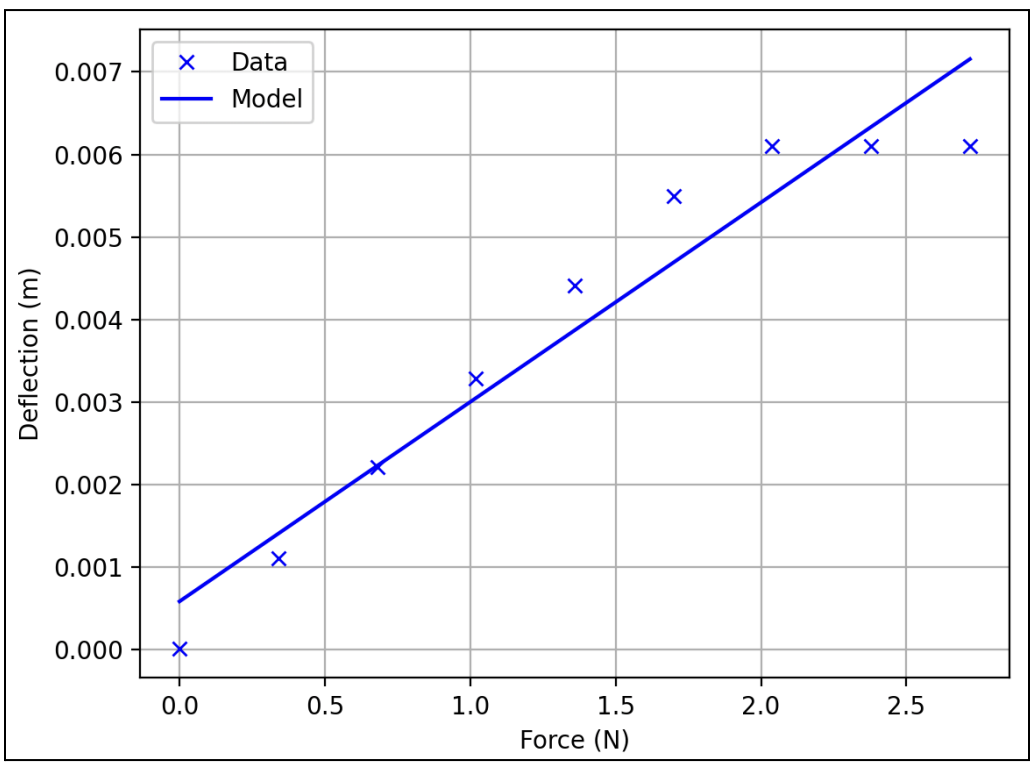
Results:

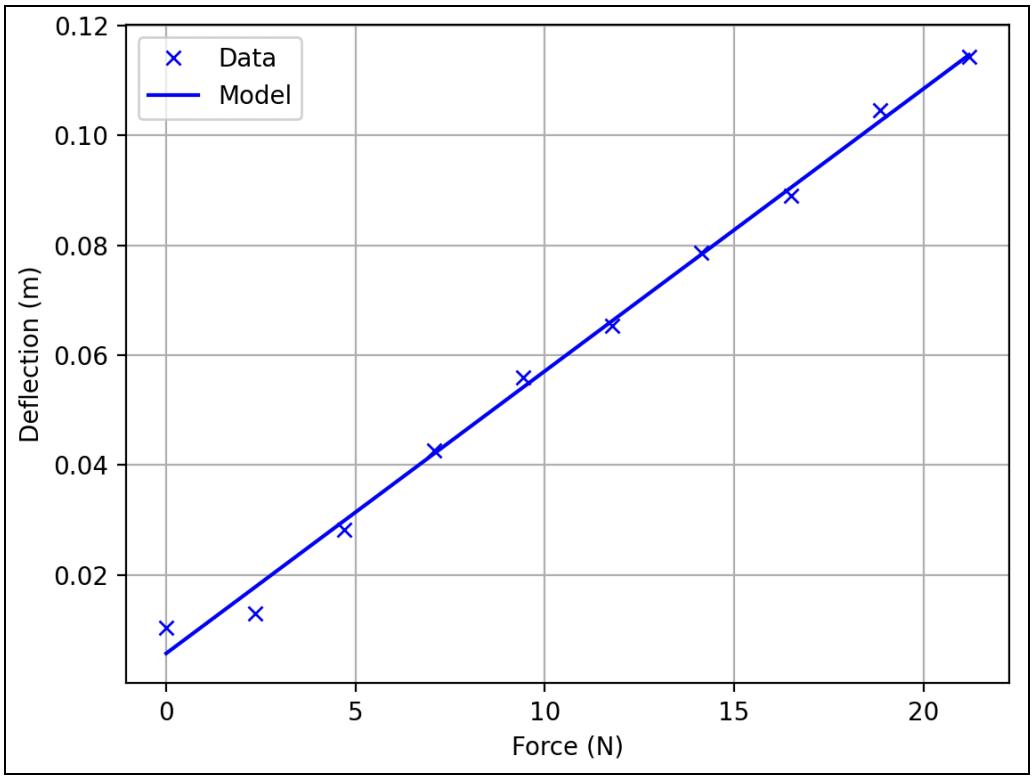
Based on the data of all 4 of the graphs, a line of best fit was generated using python. Using the equation of the best fit, the compliance and the initial deflection were calculated, which are shown in table 1 below:

Data File	Compliance (m/N)	Deflection (m)
beam1.dat	2.4163e-03	5.8703e-04
beam2.dat	5.1369e-03	5.7334e-03
beam3.dat	4.8049e-03	-2.1624e-02
cantilever.dat	3.3578e-03	5.4645e-04

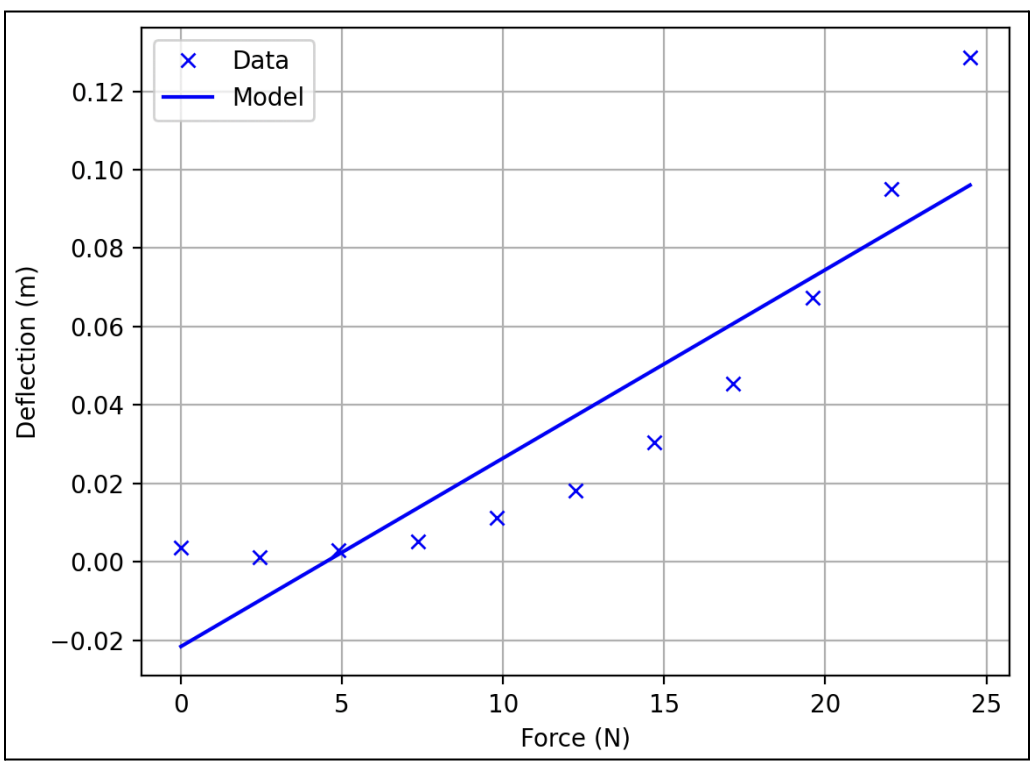
Below are the graphs of the 'Force' measured in newtons (x–axis) against the 'Deflection' measured in inches (y–axis). The graphs for all 4 data files are given in order:



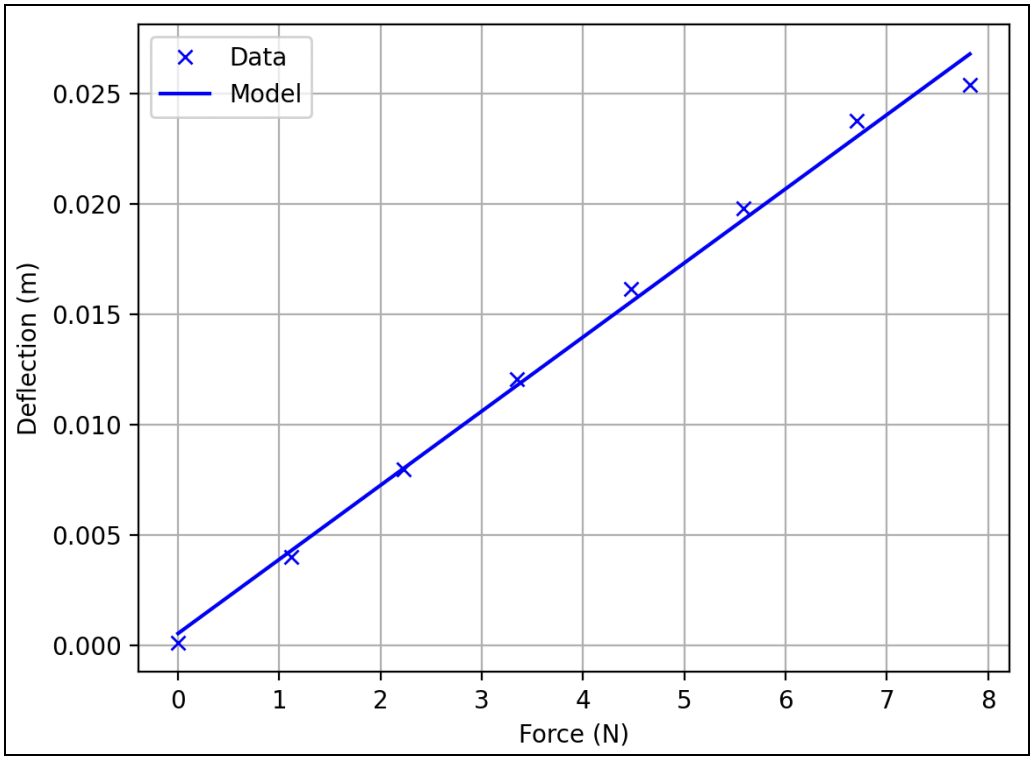
Graph 1: Force vs Deflection graph for file 'beam1.dat'



Graph 2: Force vs Deflection graph for file 'beam2.dat'



Graph 3: Force vs Deflection graph for file 'beam3.dat'



Graph 4: Force vs Deflection graph for file 'cantilever.dat'

Discussion:

The polynomial equation for the deflection as a function of force is:

$$\Delta x = (\frac{1}{k})F + (\Delta x)_0$$

Where $\frac{1}{k}$ is the spring's compliance and $(\Delta x)_0$ is the initial displacement of the end of the spring.

Substituting the compliance and initial displacement values calculated from plotting from beam1.dat, we get the below equation:

$$\Delta x = (2.4163 \times 10^{-3})\, F + 5.8703 \times 10^{-4}$$

This equation fits the plot points well, until the last few, where the plot points start approaching a constant deflection value, while the linear best fit line shows increasing deflection at a constant rate. This indicates that Beam 1 behaves like a spring under small deflections, but does not follow the same behavior under larger deflections.

Substituting the compliance and initial displacement values calculated from plotting from beam2.dat, we get the below equation:

$$\Delta x = (5.1369 \times 10^{-3})\, F + 5.7334 \times 10^{-3}$$

The equation of the best fit line fits the plot points extremely well throughout, showing that the deflection follows a linear trend with the force throughout. This indicates that Beam 2 behaves like a spring under not only small but also larger deflections as well.

Substituting the compliance and initial displacement values calculated from plotting from beam3.dat, we get the below equation:

$$\Delta x = (4.8049 \times 10^{-3})\, F - 2.1624 \times 10^{-2}$$

The equation of the best fit line does not fit the plot points well. The plot points show an almost negligible change in deflection initially, with the change increasing exponentially later on. This is not consistent with the behavior of the linear best fit line, which increases constantly throughout. This indicated that Beam 3 does not behave like a spring at neither small deflections nor larger ones. Additionally, the beam shows that it has negative initial displacement, which is not physically possible.

Substituting the compliance and initial displacement values calculated from plotting from cantilever.dat, we get the below equation:

$$\Delta x = (3.3578 \times 10^{-3})\, F + 5.4645 \times 10^{-4}$$

The equation of the best fit line fits the plot points extremely well throughout, showing that the deflection follows a linear trend with the force throughout. This indicates that the Cantilever beam behaves like a spring under not only small but also larger deflections as well.

Overall, the results indicate that almost all beams behave approximately according to Hooke's Law under small deflections, but deviations occur at higher forces. This could be because of geometric or material differences that become more and more significant as the deflection increases.

Reflection:

Before writing my code I went through the entire ReadMe document and noted down important steps in the process of my code. Next, I wrote my code in the form of an algorithm on VS Code. This helped me develop a systematic format for my code before writing it so all that would remain was transferring each line into Python code. This ensured an overall smooth process.

When I submitted my first version of the code to Gradescope I received a 1.0 out of 8.0 from the autograder, which pointed out several of my mistakes. By going through each one I corrected my code, submitting several versions and slowly working my way up to an 8.0 out of 8.0 from the autograder. One challenge was that my results were not matching the results I was supposed to get. After going through the ReadMe again and discussing with a friend, I realized that the data given to us was in inches, which we had to convert to meters. After doing this my results matched perfectly.

While going through the ReadMe I learned the basics of Hooke's Law and how the spring displacement is theoretically directly proportional to the force applied on it. I researched it further and watched a video showing how cantilever beams perform a similar motion. This was instrumental not only for the code but also for writing the report, as it allowed me to discuss my results efficiently. I also improved my skills

with Matplotlib and learned how to find polynomial best fit lines for data points. This was very useful for writing my code and helped me debug and fix errors quickly.

In Mechanical Engineering, visualizing and modeling experimental data could help improve performance of mechanical systems. For example, by analyzing the stress or pressure on a rotating system, we can predict imbalance or failure in the mechanism. Modeling this data allows us to test different designs, different conditions, or maintenance techniques without physically changing the system. This can make the process more efficient, safer, and increase performance.

Appendix: Python Code

```
import numpy as np
import matplotlib.pyplot as plt

def analyze_plot_data_file(filename):
	"""
	Reads data from a cantilever beam experiment, fits a linear model to
	find the compliance and initial deflection, plots deflection vs. force,
	saves the figure as a PNG file with the same base name as the data file,
	and returns an ndarray of the model parameters.
	"""
	data = np.loadtxt(filename)
	mass = data[:, 0]
	deflection = data[:, 1] * 0.0254
	force = 9.81 * mass
	best = np.polyfit(force, deflection, 1)
	compliance = best[0]
	initial_deflection = best[1]
	plot_name = filename.split(".")[0]
	plot_name = plot_name.split("/")[1]+".png"

	dmodel = np.polyval(best, force)
	plt.plot(force, deflection, "bx", label="Data")
	plt.plot(force, dmodel, "b-", label="Model")
	plt.xlabel("Force (N)")
	plt.ylabel("Deflection (in)")
	plt.grid(True)
	plt.legend()
	plt.show()
	plt.savefig(plot_name)
	plt.close()

	return np.array([compliance, initial_deflection])

def main():
	for filename in ['data/beam1.dat', 'data/beam2.dat', 'data/beam3.dat', 'data/cantilever.dat']:
		parameters = analyze_plot_data_file(filename)
		compliance, initial_deflection = parameters
		shortname = filename.split('/')[-1]
		print(f"{shortname}: {compliance:.4e}, {initial_deflection:.4e}")

if __name__ == "__main__":
	main()
```